

Takeaways & Preview

- **Gradient flows in Wasserstein space**
 - $\sim \tau \rightarrow 0$ limit of gradient descent
 - JKO scheme: variational time-stepping with Wasserstein proximity.
 - PDE form: $\partial_t \rho = \nabla \cdot \left(\rho \nabla \frac{\delta \mathcal{E}}{\delta \rho} \right)$
 - Applications: diffusion models, implicit PDE solvers, physics-inspired ML.
- **Schrödinger bridges** (stochastic view)
 - Path-space KL minimization relative to Brownian motion.
 - Equivalent to entropic OT in the static case.
 - Connects OT to diffusion processes, score-based generative models
- **Next time:** OT vs other statistical divergences (KL, IPMs, etc)

